Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Secure maritime digital twin platform for electrified vessel energy management, safety analytics, and compliance reporting			
Korea's 12 National Strategic Technology Fields	Semiconductors & Displays ( ), Secondary Batteries ( ), Advanced Mobility ( ), Next-generation Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (X), Hydrogen ( ), Cyber security (X), Artificial Intelligence (X), Next-generation Communications ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communications		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Software	SW Solutions	Batteries	Secondary Batteries
Concept of Proposed Technology	**Digital twins in the maritime sector** A digital twin is a *real-time, data-synchronized virtual replica* of a physical object, system, or environment. It integrates sensor data, models, and analytics to mirror the behaviour and condition of its physical counterpart throughout its lifecycle. True digital twins continuously update with live data from the physical asset, enabling simulation, monitoring, prediction, and optimization; this distinguishes them from static models or digital shadows. In the maritime industry, a digital twin typically represents a *vessel, fleet, port infrastructure, or navigable maritime environment* in a virtual space. The twin reflects physical characteristics and operational states in near real time using data from onboard sensors, Internet of Things (IoT)			

devices, satellite feeds, and other sources.

Digital twin technology has been adopted at multiple stages of the maritime value chain:

- Design and construction: Digital replicas of ships enable exploration of design choices and structural behaviours without physical prototypes, reducing risk and development costs.
- Operational management: Connected twins allow fleet operators to *monitor performance, estimate fuel consumption, and simulate scenarios* such as equipment failure, weather impacts, or emergency responses in a virtual environment.
- Maintenance and lifecycle planning: Using live data and predictive analytics, digital twins can forecast equipment failures before they occur and support scheduling of maintenance to *minimize downtime and extend asset life*.
- Safety and training: Digital twins offer immersive, risk-free training environments and hazard simulations for crew and port staff.
- Smart ports and waters: Beyond individual vessels, digital twin frameworks are emerging for *port operations* and *navigable water systems*, enabling real-time visualization of ship movements, environmental conditions, and operational logistics.

Major maritime hubs are actively deploying digital twins. For example, the Port of Singapore launched a *Maritime Digital Twin* to demonstrate real-time vessel monitoring and underwater hull inspection use cases.

Digital twin technology in maritime is enabled by advances in IoT, big data analytics, artificial intelligence, and high-fidelity simulation. These technologies allow the virtual representation to evolve alongside the physical system, improving decision-making, operational efficiency, environmental performance, and safety

This proposal targets a secure, model-based digital twin solution for maritime vessels that combines:

- Hybrid and electric powertrain and battery digital twins (electro-thermal behavior, degradation, and energy flow)
- Voyage and operational context (route, speed profile, weather/sea-state proxies, auxiliary loads)
- Safety and anomaly analytics (early detection of abnormal operating patterns, risk indicators, event logging)

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	- Interoperable data sharing and governance (trusted data exchange between vessel owners, OEMs, service providers, and authorities) The envisioned outcome is an integrated "vessel energy and safety digital twin" that supports design-time optimization and in-operation decision support, while keeping data sovereignty with the data owner (ship operator).
Rationale for Joint R&D Project	Electrification and hybridization of maritime vessels increase system complexity and introduce new safety and performance challenges, especially under variable missions and harsh environments. Korean SMEs building maritime digital-twin and navigation-data services can accelerate commercialization by coupling their operational platform with TNO's physics-based powertrain and battery modelling, control-oriented digital twins, and secure data-space connectors. Joint R&D is required to: - Combine domain-specific maritime operational data and platform integration (Korean SME) with validated multi-physics models and test-backed parameter identification (TNO). - Establish a trusted, auditable data pipeline and governance model needed for cross-company collaboration and future digital product passport style reporting (in line with data-space architectures).
Key Words	Maritime digital twin; electrified vessels; energy management; anomaly detection; secure data sharing

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	Develop and demonstrate a secure maritime digital twin stack that enables: 1) Predictive energy management and range estimation for hybrid and electric vessels under mission profiles. 2) Battery and powertrain state estimation (SoC, SoH, temperature) and degradation forecasting. 3) Safety analytics, including anomaly detection and event logging for critical conditions. 4) Standardized data interfaces and access control to support multi-stakeholder data sharing without loss of data sovereignty.			
R&D Target Specifications	Target specifications (to be finalized with the Korean SME and pilot vessel operators during Phase 1): - Energy prediction accuracy: <= 5% mean absolute percentage error (MAPE) on voyage energy consumption for representative missions (benchmarked against logged data), once operational data covering at least three distinct sea-state levels are available for model development and validation. - SoC estimation error: <= 2% absolute error over a mission profile (validated with reference measurements where available). - Degradation forecasting: capability to project capacity fade and resistance growth trends over >= 3 months of operational data, with documented uncertainty bounds. - Anomaly detection: detection of predefined abnormal patterns (e.g., abnormal temperature rise, voltage instability, power limitation events) with >= 90% recall at <= 5% false positive rate in pilot logs. - Secure data exchange: implementation of authenticated, policy-based access control and auditable event logging for all data transfers between stakeholders.			
R&D Plan	Phase 1 (Pre-planning / pre-research, 6 months) Task 1. System definition and data model - Define use cases (energy prediction, safety analytics, reporting) and minimum viable dataset. - Map operational signals (navigation, power, battery, environment proxies) to a joint data model. Task 2. Digital twin baseline models and identification strategy - Select model fidelity: control-oriented reduced-order models for real-time use plus offline high-fidelity reference models. - Define parameter identification methods based on available operational data and targeted tests. Task 3. Secure data pipeline and governance - Implement a secure connector architecture and access control policies for data ingestion and sharing. - Define audit trails, event schema, and data-quality checks. Deliverables: architecture, data specification, baseline models, pilot plan, KPI definition. Phase 2 (Main R&D, 24-30 months) Task 4. Hybrid/electric vessel powertrain and battery digital twin development (TNO lead) - Electro-thermal battery models and degradation sub-models.			

	- Powertrain and auxiliary load models tied to mission profiles. - Real-time estimators (SoC, SoH, thermal state) and predictive controllers for energy management. Task 5. Safety and anomaly analytics (joint) - Feature engineering from time-series operational signals. - Hybrid physics-informed and data-driven anomaly detection and event classification. - Explainability layer to support operator trust and maintenance actions. Task 6. Platform integration and pilot validation (Korean SME lead) - Integrate the digital twin services into the SME platform (cloud/edge as appropriate). - Pilot with operational datasets and, where possible, controlled experiments. Task 7. Demonstration and exploitation package (joint) - Documented APIs, model documentation, KPI results, and commercialization roadmap. - IP strategy and publication plan. Milestones: M1 (month 6): Phase 1 package complete (architecture, data model, baseline models, connector prototype). M2 (month 12): First integrated digital twin service with offline validation. M3 (month 18): Real-time estimator and anomaly analytics beta on pilot datasets. M4 (month 30): Field demonstration with KPI evidence and commercialization plan.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO (Research Institute) responsibilities - Develop physics-based and control-oriented digital twin models for hybrid/electric vessel powertrains and batteries (electro-thermal, degradation). - Design state estimation and predictive energy management algorithms. - Provide secure data-sharing architecture guidance (data spaces, connectors, access control, auditability) and implement prototype connector components as needed. - Support validation methodology, KPI definition, and uncertainty quantification. - Lead scientific dissemination and contribute to IP generation (jointly with the SME, per project agreement).
Expected R&R of Korean SME Partner	- Act as the **primary maritime digital platform owner and system integrator**, taking full responsibility for the operational deployment of the maritime digital twin services. - Define **maritime operational requirements and service use-cases** based on partner's navigation, fleet performance monitoring, and decarbonization solutions. - Provide and manage **high-frequency operational datasets**, including navigation data, vessel motion, power and energy system signals, and contextual voyage information, under agreed data-governance and security policies. - Lead **platform integration** of TNO's digital twin models and analytics into partner's cloud and edge architecture, including API orchestration, data pipelines, and service lifecycle management. - Design and implement **user-facing applications and dashboards**, translating digital-twin outputs into actionable insights for ship operators, fleet managers, and compliance stakeholders. - Coordinate **pilot vessel selection, onboard deployment, and field validation**, including feedback loops for model tuning and service refinement. - Lead **commercialization and go-to-market activities** in Korea and selected international markets, leveraging partner's existing customer base and maritime partnerships. - Contribute to **joint IP creation, exploitation strategy, and standardization alignment**, ensuring compatibility with emerging maritime digital standards and regulatory frameworks.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected outcomes - A validated digital twin service stack for electrified vessels, enabling improved energy efficiency, safer operation, and reduced downtime through predictive analytics. - A secure and interoperable data-exchange approach that can scale to multi-stakeholder maritime ecosystems. Impact - Commercial: Accelerated time-to-market for the Korean SME's maritime platform through integration of high-fidelity models and safety analytics; differentiated product offering for international customers. - Technical: Demonstrated linkage between mission profiles and powertrain/battery behavior, supporting better design, operation, and maintenance decisions. - Strategic: Supports the broader transition to low-emission maritime transport by enabling reliable electrification and trusted data collaboration.

6. References

[1] Wilkins, S., Shipurkar, U., Dadikozyan, A., Veldhuis, C. "Co-Design and Energy Management for Future Vessels" (MOSES 2023). DOI: https://doi.org/10.59490/moses.2023.666 (accessed 2025-12-18).

[2] Rath, S., Wilkins, S. "Prediction of Electric Vehicle Charge Profile using Battery Digital Twin" (conference paper, 2024). TNO publication record: https://resolver.tno.nl/uuid:a4b05a75-0e61-41a7-8368-f006653ff414 (accessed 2025-12-18).

[3] TNO people profile: Steven Wilkins. https://www.tno.nl/en/about-tno/our-people/steven-wilkins/ (accessed 2025-12-18).

[4] TNO Facility: Battery Lab. https://www.tno.nl/en/technology-science/facilities/battery-lab/ (accessed 2025-12-18).

[5] TNO unit page: TNO research on powertrains. https://www.tno.nl/en/about-tno/organisation/units/mobility-built-environment/tno-research-powertrains/ (accessed 2025-12-18).

[6] Battery Competence Cluster NL news: "A passport to serve the battery circular economy" (18 Jul 2023). https://batterycompetencecluster.nl/en/news/a-passport-to-serve-the-battery-circular-economy (accessed 2025-12-18).

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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TNO Battery Modeling, Diagnostics, and Digital Twin Capabilities

TNO has long-standing expertise in physics-based battery modeling, diagnostics, and digital twin development across cell, module, and pack levels. Within the OptiBatt program, TNO develops coupled electrochemical–thermal–aging models that support performance prediction, lifetime assessment, and safety analysis under real-world operating conditions. These models are systematically validated using TNO's battery testing laboratories and climate-controlled test facilities, enabling traceable and industrially deployable digital twins.

Key elements include:

- Multi-physics battery models capturing electrochemistry, heat generation, and degradation.
- Model-based diagnostics for state-of-health (SoH) and remaining useful life (RUL).
- Digital twin frameworks linking models with operational data for predictive control and safety assurance.

Publicly citable sources include:

• TNO OptiBatt – Optimal Batteries roadmap (TNO, 2024): https://www.tno.nl/en/focus-areas/sustainable-vehicles/roadmaps/optimal-batteries/

• TNO Battery Labs and testing infrastructure (TNO, 2024): https://www.tno.nl/en/focus-areas/sustainable-vehicles/labs/battery-labs/